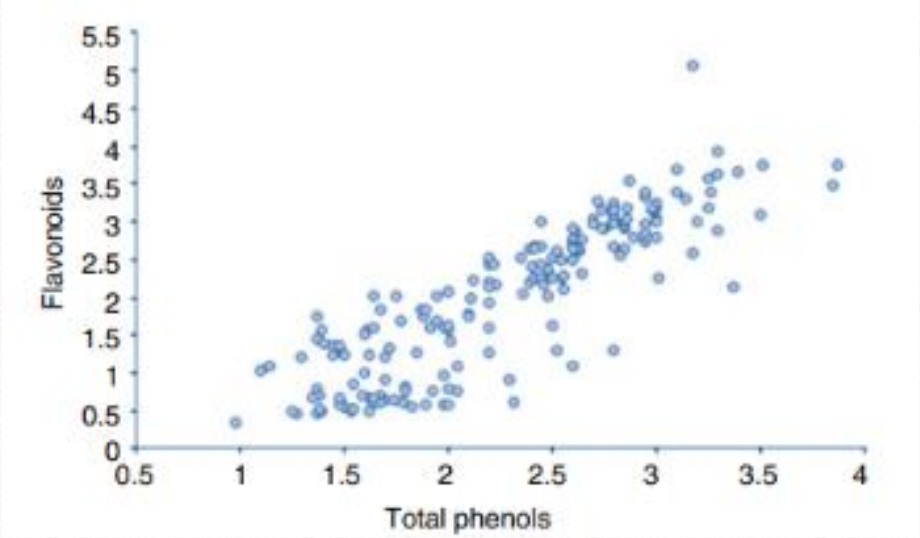
Understanding Relationships

Visualizing Relationships

- Is there a relationship between
  - Interest Rate and Inflation
  - Education level and Income
- Different ways of understanding the relationship
  - Scatterplots
  - Summary Tables and Charts
  - Specific Calculated metrics

Scatter Plot

- A type of graph that shows the relationship between two continuous variables measured on ratio/interval scales
- Each point represents one observation

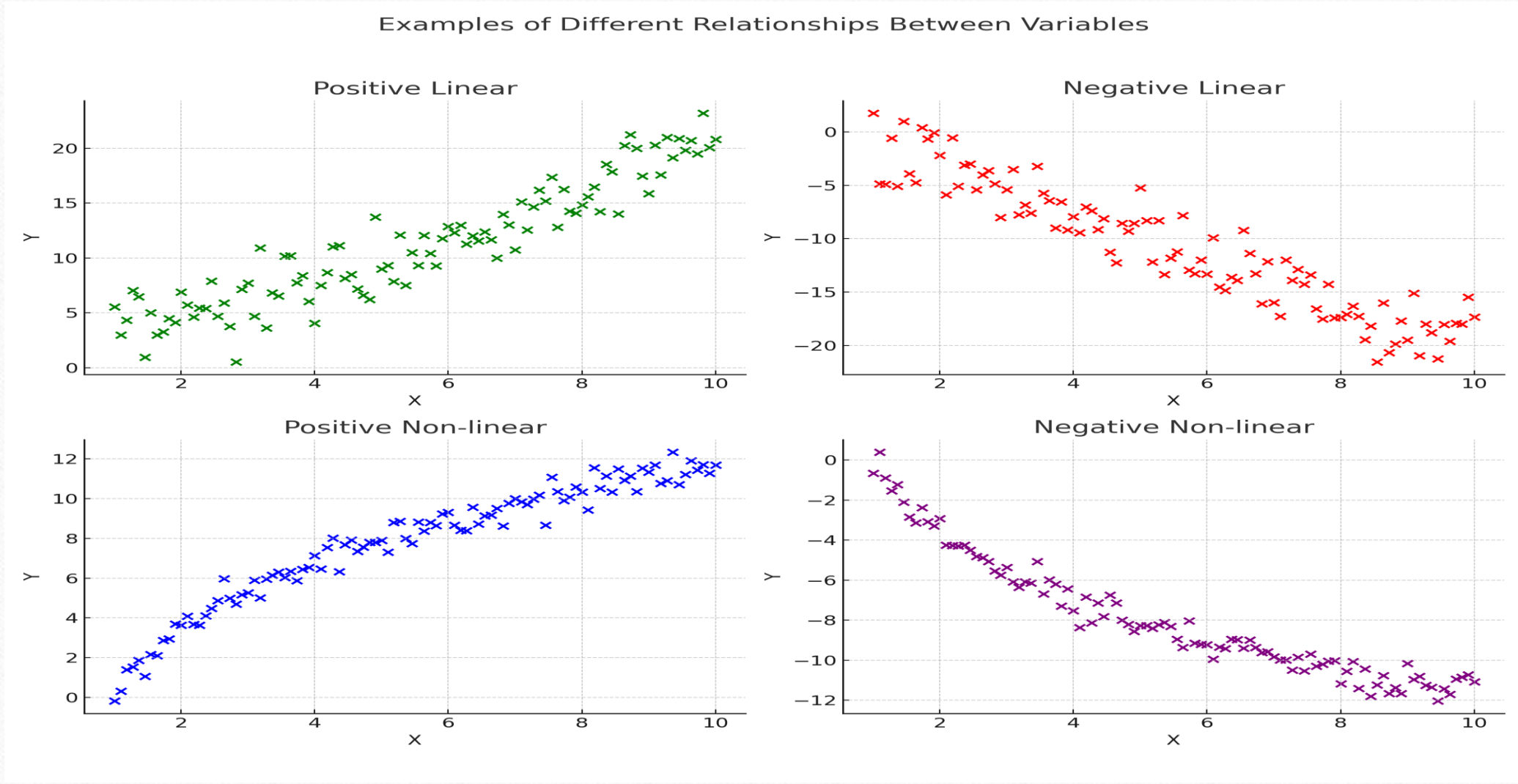


Scatter Plot

- It allows you to see the type of relationship between the variables
  - Positive – Higher values in the first variable coincide with higher values in the second variable and lower values in the first variable coincide with lower values in the second variable.
  - Negative – Higher values in the first variable coincide with lower values in the second variable and lower values in the first variable coincide with the higher values in the second variable.

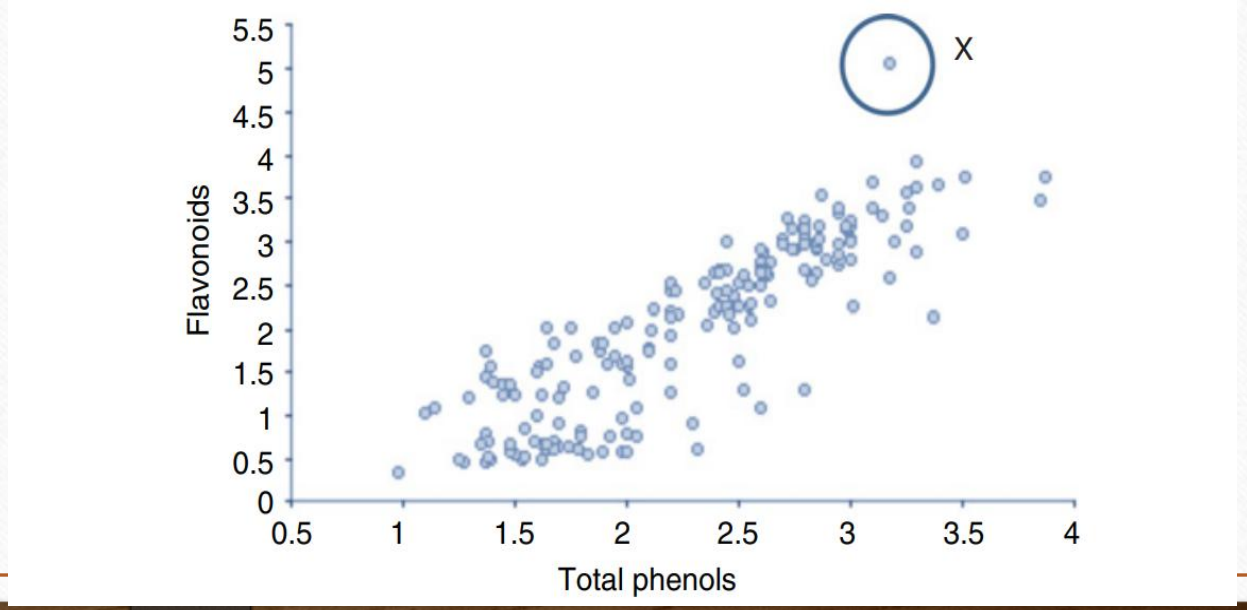
Scatter Plot

- Based on the nature of the relationships
  - Linear Relationship – When the second variable changes proportionally in response to the changes in the first variable.
  - Nonlinear Relationship – When the changes in the first variable is not proportional to the change in the second variable.


Examples of Different Relationships Between Variables
Positive Linear
Negative Linear
Positive Non-linear
Negative Non-linear
X
Y

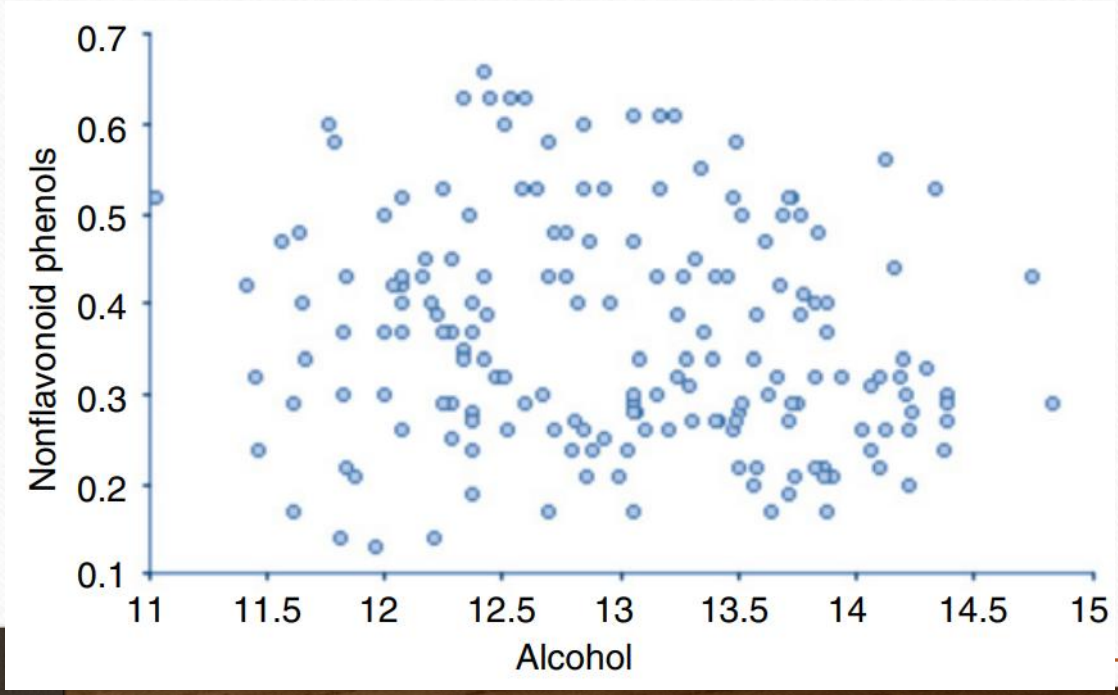
Scatter Plot

- A scatterplot can also show if there are points that do not follow this linear relationship. These are referred to as outliers.



Scatter Plot

- Scatterplots can also show the lack of any relationship.



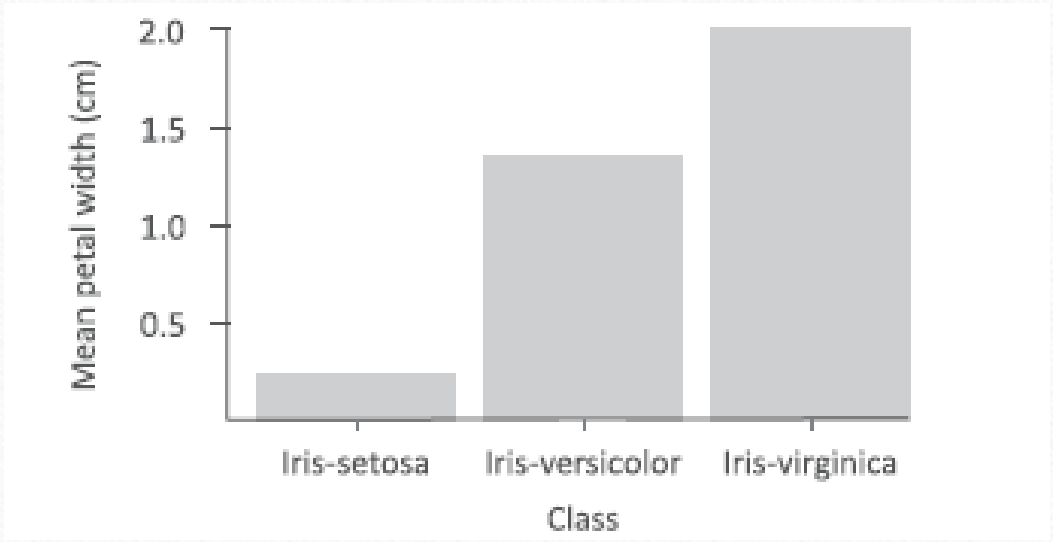
Summary Tables and Charts

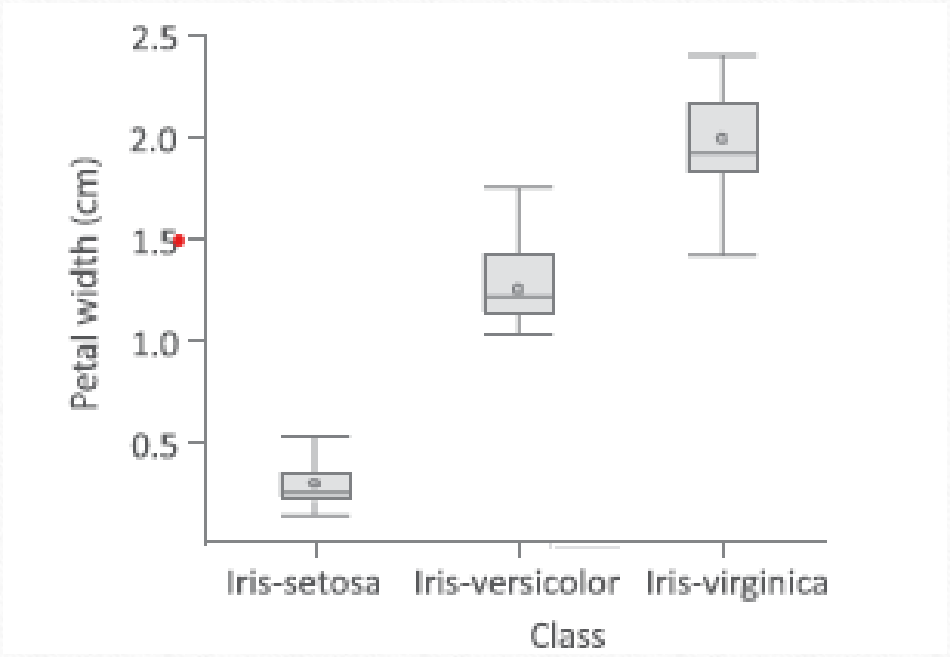
- A simple way to understand relationships between variables
- Especially useful when one variable is categorical or discrete
  - Rows = Groups (categories or bins of continuous variables)
  - Columns = Summary statistics of the second variable
- Examples of stats used: Mean, Median, Min, Max, Std Dev, Count

Example-Relationship between class and petal width (cm) (Nominal)

Class	Count	Min (Petal width)	Max (Petal width)	Mean (Petal width)	Median (Petal width)	SD (Petal width)
Iris-setosa	50	0.1	0.6	0.244	0.2	0.107
Iris-Versicolor	50	1	1.8	1.33	1.3	0.198
Iris-Virginica	50	1.4	2.5	2.03	2	0.275

Example-Relationship between class and petal width (cm)





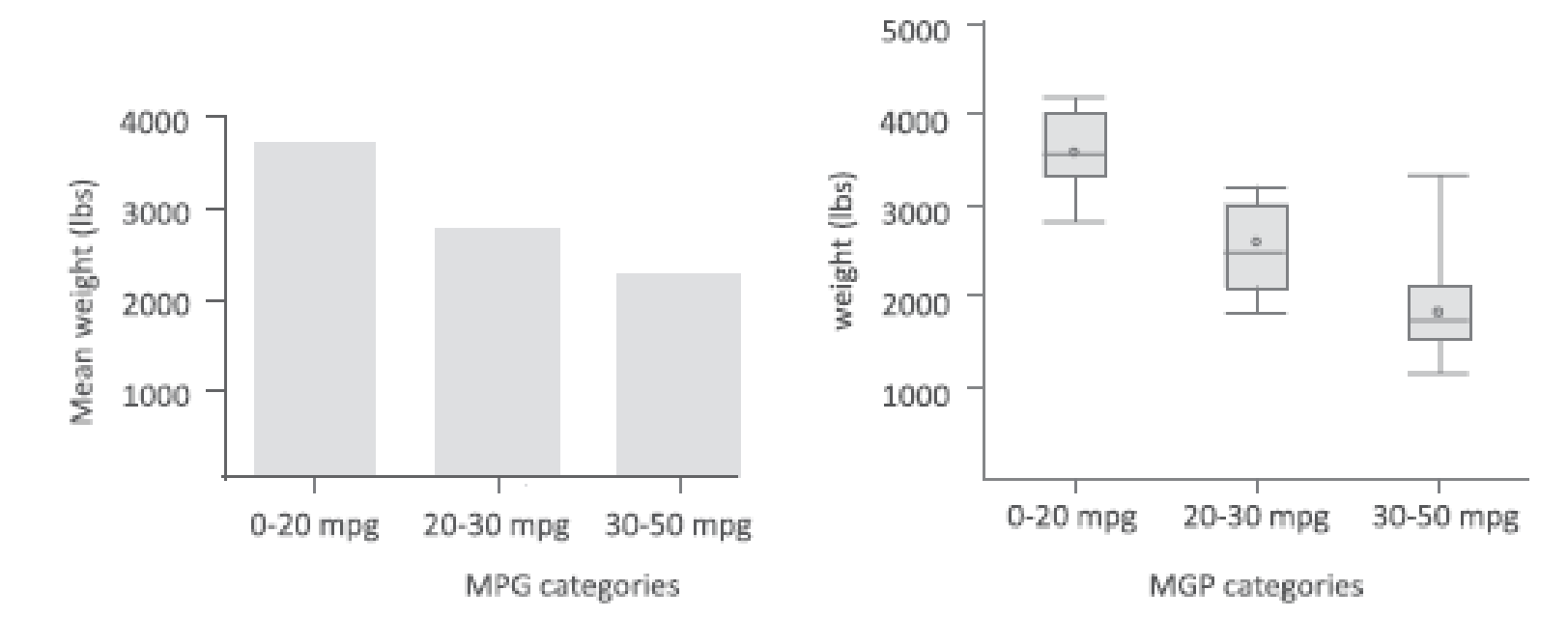
- **Overall Relationship**
- From both charts, we can conclude that:
- There is a **strong relationship between Iris species and petal width**.

- The average petal width follows the order:
  Setosa < Versicolor < Virginica.
- Petal width is a **good discriminating feature** for classifying Iris species.
- **Setosa** has the smallest and most consistent petal widths.
- **Virginica** has the largest petal widths and the greatest variation.
- **Versicolor** lies between the other two species in both mean and variability.
- While **Setosa** can be clearly separated using petal width, **Versicolor** and **Virginica** show slight overlap, indicating that additional features (such as petal length or sepal measurements) may improve classification accuracy.

Summary Table -Ordinal

MPG categories	Count	Mean (weight)
0–20 MPG	151	3832
20–30 MPG	151	2612
30–50 MPG	90	2157

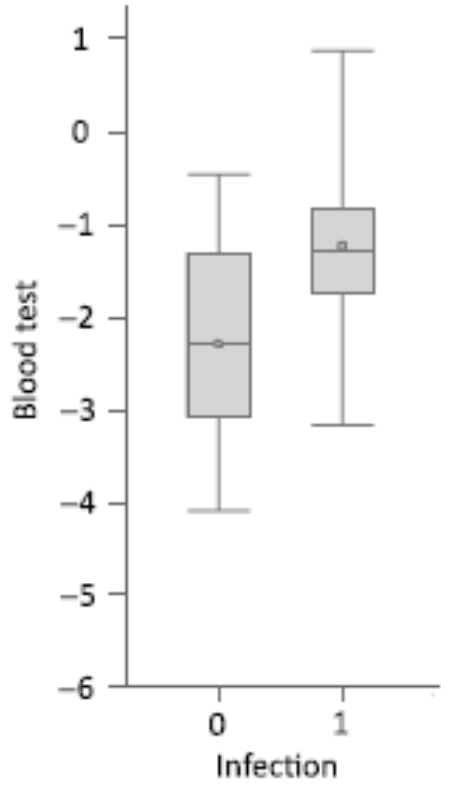
Summary Table -Ordinal



- From both charts, we can conclude:
- There is a **strong inverse relationship** between vehicle weight and fuel efficiency.
- As vehicle weight increases, fuel efficiency generally decreases.
- Vehicles in the **0–20 mpg** category are the heaviest on average.
- Vehicles in the **30–50 mpg** category are the lightest on average.
- The **20–30 mpg** category serves as an intermediate group with a wider range of vehicle weights.
- The overlap in box plots indicates that **weight is an important predictor of MPG, but not the only factor** affecting fuel efficiency.

Binary Variable – Continuous data

- 1/0- Patient has infection/no infection
- Using Blood Test – calculates within -5.0 to 1.0
- Higher value – has infection and vice versa



Infection	Count	Minimum (blood test)	Maximum (blood test)	Mean (blood test)	Standard deviation (blood test)
0	23	−4.13	−0.393	−2.18	1.04
1	32	−3.05	0.893	−1.09	0.834

- **Relationship Between Infection and Blood Test**
- The relationship can be summarized as:
- Infection is associated with **higher blood test values**.
- Non-infected patients have **lower average blood test values**.
- Non-infected patients show **greater variability**.
- There is **substantial overlap** between the two groups.

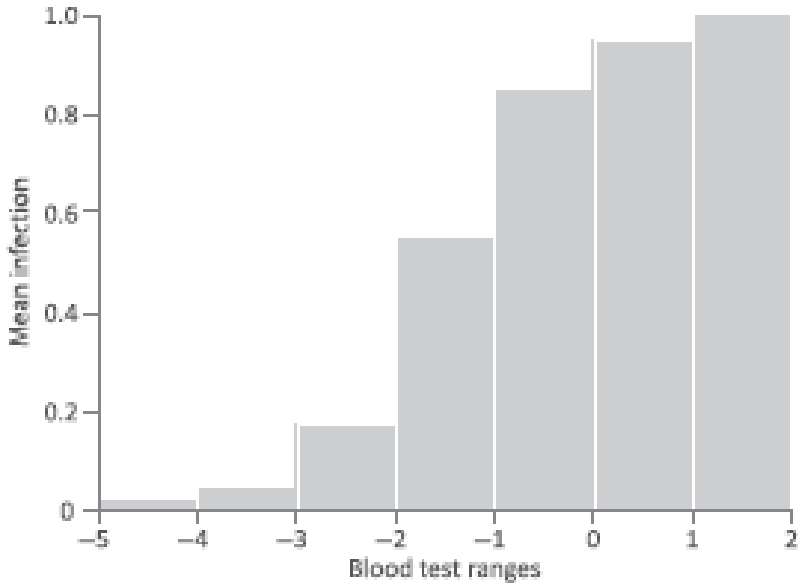
Overall Conclusion

- From the summary table and box plot, we conclude that:
- Patients **with infection (1)** tend to have **higher blood test values** than those **without infection (0)**.
- The **mean increases** from **−2.18** (no infection) to **−1.09** (infection), suggesting a positive association between infection status and the blood test measurement.
- The **non-infected group** exhibits greater variability, as reflected by the higher standard deviation and wider box plot.
- Although the infected group generally has higher values, the distributions overlap considerably, indicating that **blood test values alone are not sufficient to distinguish infected from non-infected patients with complete accuracy**.

Binary Variable – Continuous data

- Blood tests results are continuous – organize the blood test sample in ranges

Blood test ranges	Count	Mean (Infection)
–5 – –4	2	0.02
–4 – –3	6	0.06
–3 – –2	9	0.19
–2 – –1	15	0.55
–1 – 0	11	0.85
0 – 1	9	0.95
1 – 2	3	1.0



Practical Interpretation

- Suppose a new patient's blood test result is:
  - **-4.5** → Very low chance of infection (~2%).
  - **-2.5** → Moderate chance (~19%).
  - **-1.5** → Around a **55%** chance of infection.
  - **-0.5** → High chance (~85%).
  - **0.8** → Very high chance (~95%).
- Thus, the blood test value can help estimate the likelihood of infection, although clinical decisions should also consider symptoms and other diagnostic tests.

Overall Conclusion

- From the table and chart, we conclude that:
- There is a **strong positive relationship** between blood test values and infection probability.
- Patients with **lower blood test values** are much less likely to have an infection.
- As blood test values increase, the **proportion of infected patients increases steadily**.
- The probability rises from **2%** in the lowest blood test range to **100%** in the highest range.
- The marked increase around the **-2 to -1** interval suggests a possible threshold where the risk of infection becomes significantly higher.

Cross-Classification Tables

- Cross-classification/Contingency tables – provide insights into relationship between two categorical variables
- E.g. Gender vs Product Purchased or Gender vs Product preference
- Contingency tables can represent variables with more than two values
  - Test Results and Infection Class for series of patients
  - Test Results – Blood Test Positive / Negative
  - Infection Class – Infection Positive / Negative

Cross-Classification Tables - Nominal

- Total Observations – 55, with 27 with Blood test –ve and 28 +ve
- 23 with infection –ve and 32 +ve, center cells correspond to patients with pairs of values
- 22 and 17 perfectly classified while 10 and 6 incorrectly classified

		Infection Class		
		Infection negative	Infection positive	Totals
Test Results	Blood test negative	17	10	27
	Blood test positive	6	22	28
	Totals	23	32	55

Identify the Four Outcomes

Outcome	Value	Meaning
True Positive (TP)	22	Test correctly identified infected patients.
True Negative (TN)	17	Test correctly identified non-infected patients.
False Positive (FP)	6	Test indicated infection, but the patient was not infected.
False Negative (FN)	10	Test missed an infection (patient was infected but test was negative).

Interpretation

- **a) Correct Diagnoses**
- The blood test correctly diagnosed:
  - **22 infected patients**
  - **17 non-infected patients**
- Total correct diagnoses:
  - 22+17=39
- So,
  - **39 out of 55 patients were classified correctly.**

Interpretation

- **b) Incorrect Diagnoses**
- The blood test made mistakes for:
  - **6 False Positives**
  - **10 False Negatives**
- Total mistakes:
  - 6+10=16

Interpretation

- **c) Which Error is More Common?**
- There are
  - **6 False Positives**
  - **10 False Negatives**
- The test misses infections more often than it incorrectly identifies infection.
- In medicine, **false negatives are generally more concerning**, because an infected patient may not receive timely treatment.
- The interpretation helps us to calculate performance measures of Accuracy, Precision, Recall and Specificity.

Cross-Classification Tables - Ordinal

- Gender vs age-group
- 9 categories
- Total observations 21790

		Gender		
		Male	Female	Totals
Age-group	10–19	847	810	1657
	20–29	4878	3176	8054
	30–39	6037	2576	8613
	40–49	5014	2161	7175
	50–59	3191	1227	4418
	60–69	1403	612	2015
	70–79	337	171	508
	80–89	54	24	78
	90–99	29	14	43
	Total	21,790	10,771	32,561

Overall Interpretation

- From the contingency table, we conclude that:
- The dataset consists of **32,561 individuals**, with **21,790 males** and **10,771 females**.
- **Males outnumber females** in every age group.
- The **30–39 years** age group has the highest population, followed by the **20–29** and **40–49** age groups.
- The population decreases progressively after the age of **40**, with very few individuals aged **80 years and above**.
- The data suggests that the majority of the population belongs to the **working-age group (20–49 years)**.
- The largest gender difference occurs in the **30–39 years** age group, where the number of males substantially exceeds the number of females.

Calculating metrics about relationships

- Measure the strength of the relationship between two variables
- Metrics depend on type of variables: categorical and continuous
- Metrics-
  - Correlation Coefficient
  - Kendall Tau- The below will be covered in L21, L22, L23
  - T-test
  - ANOVA
  - Chi-square

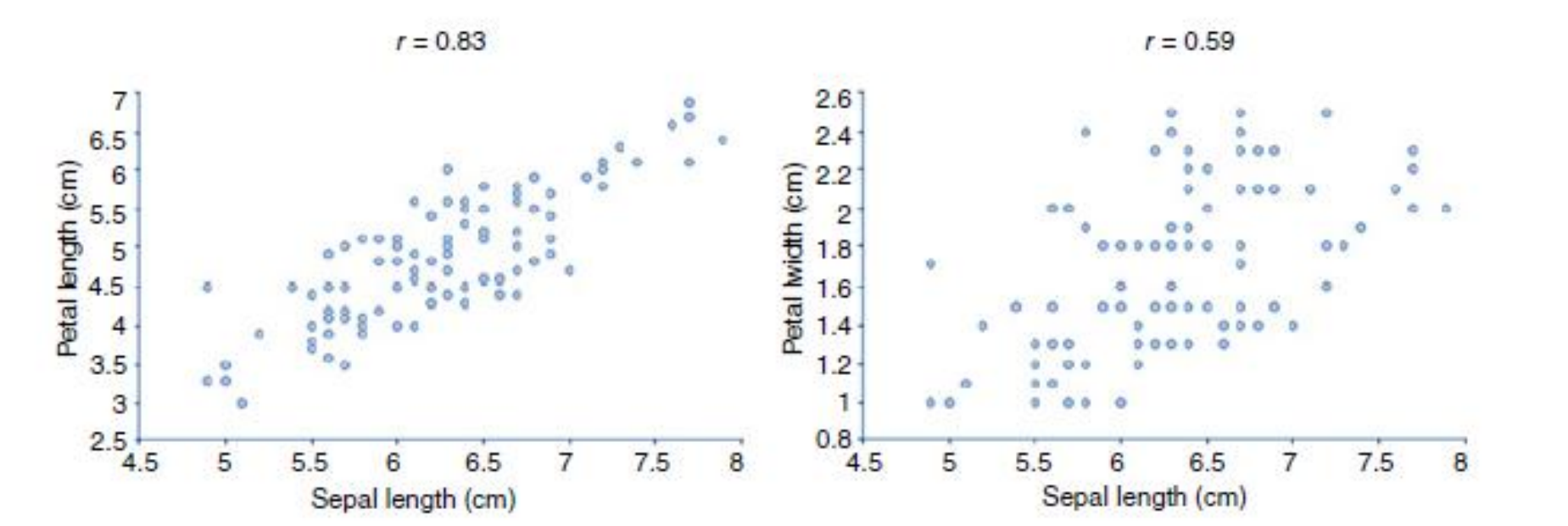
Correlation Coefficient

- For the pair of the variables measured on interval or ratio scale correlation coefficient can be calculated
- Quantifies the linear relationship between the variables (-1.0 to +1.0)

$$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$$

- The **correlation coefficient (r)** measures the **strength and direction** of the linear relationship between two variables.
  - **r = +1** → Perfect positive correlation
  - **r = 0** → No linear correlation
  - **r = -1** → Perfect negative correlation

Correlation Coefficient

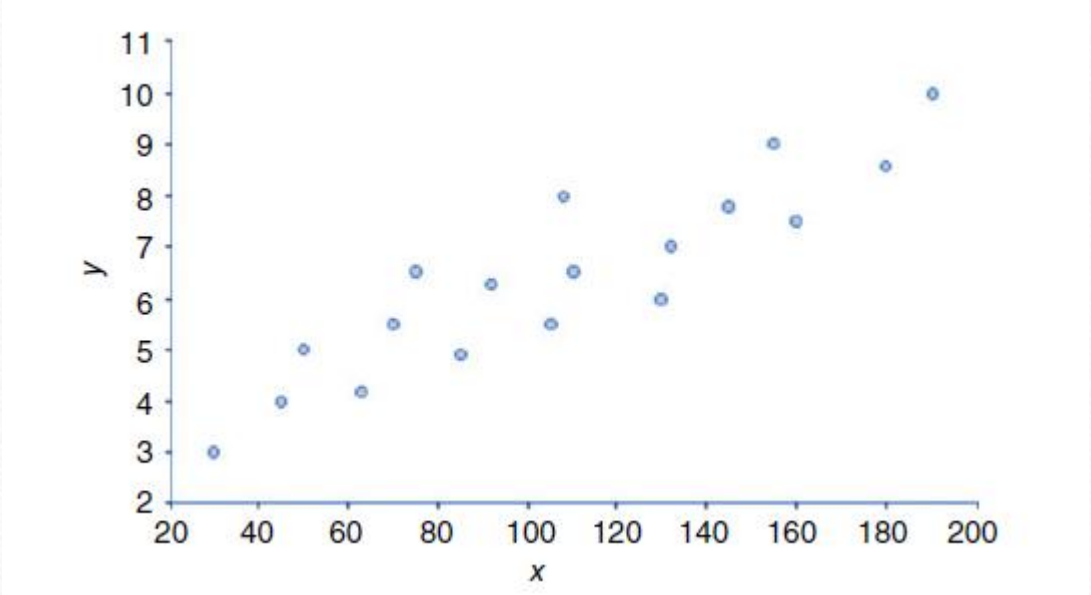


x	y
92	6.3
145	7.8
30	3.0
70	5.5
75	6.5
105	5.5
110	6.5
108	8.0
45	4.0
50	5.0
160	7.5
155	9.0
180	8.6
190	10.0
63	4.2
85	4.9
130	6
132	7

Correlation Coefficient

- Calculate s^2_x, s^2_y, s_x and s_y
- Calculate r
- What does the value of r imply?

r=0.91 showing strong association between x and y as shown in the figure



Kendall Tau

- Approach for measuring association between pairs of variables.
- It is based on ranking of the observations for 2 variables
- The ranking is derived by ordering the values and then replacing the actual values with rank 1 to n .
- Formula is based on Concordant and discordant pairs of observations

Kenall Tau Interpretation

Kendall's Tau (τ)	Interpretation
+1.00	Perfect positive association
+0.80 to +0.99	Very strong positive association
+0.60 to +0.79	Strong positive association
+0.40 to +0.59	Moderate positive association
+0.20 to +0.39	Weak positive association
0.00 to +0.19	Very weak or negligible positive association
0	No association
-0.19 to 0.00	Very weak or negligible negative association
-0.20 to -0.39	Weak negative association
-0.40 to -0.59	Moderate negative association
-0.60 to -0.79	Strong negative association
-0.80 to -0.99	Very strong negative association
-1.00	Perfect negative association

Kendall Tau

- A pair of observation is **Concordant**
  - If the difference of one of the two variable's values is in the same direction as the difference of the other variable's value.
  - Eg: the observations A and B are concordant, if X_B-X_A and Y_B-Y_A are in the same direction for variables X and Y. (We get the same difference if we do (X_A – X_B)
- A pair of observation is **Discordant**
  - If the difference of two variable's values move in the opposite direction
  - Eg: the observations A and B are concordant, if X_A-X_B is positive and Y_A-Y_B is negative – the move in the opposite direction. (We get the same difference if we do (Y_B – Y_A)

Kendall Tau

- A final Kendall Tau measure is computed based on these computed sums
- Kendall Tau measures associations between variables with 1 indicating a perfect ranking and -1 a perfect disagreement of the rankings
- Zero values – assigned when the ranks are tied – indicates a lack of association i.e the two variables are independent
- Variables n_c – no of concordant and n_d – no of discordant pairs

$$\tau_A = \frac{n_c - n_d}{n(n-1)/2}$$

Kendall Tau

TABLE 4.3 Data Table of Rankings for Two Variables

Observation name	Variable X	Variable Y
A	1	2
B	2	4
C	3	1
D	4	3
E	5	6
F	6	5
G	7	7
H	8	8
I	9	10
J	10	9

TABLE 4.4 Calculation of Concordant and Discordant Pairs

Observation Name	Variable X	Variable Y	Concordant Pairs	Discordant Pairs
A	1	2	8	1
B	2	4	6	2
C	3	1	7	0
D	4	3	6	0
E	5	6	4	1
F	6	5	4	0
G	7	7	2	1
H	8	8	2	0
I	9	10	0	1
J	10	9	0	0
		Sum:	39	6

Kendall Tau

- The value of Tau A is 0.73
- Positive Association: Since τ_a is positive, higher ranks in X tend to correspond to higher ranks in Y.
- Values of τ between 0.7 and 0.9 are generally considered strong.
- Here, $\tau_a \approx 0.73$ indicates a substantial agreement between the two rank orderings.

Kendall Tau (with ties)

- In case of ties based on either variable, the formula Tau B is used
- It considers the ties in the first variable(t_x) and ties in the second variable (t_y)
- Uses the following formula

$$\tau_B = \frac{n_c - n_d}{\sqrt{(n_c + n_d + t_x)(n_c + n_d + t_y)}}$$

Kendall Tau (with ties)

Obs	X	Y
A	1	1
B	2	1
C	2	2
D	3	3
E	4	3

$$\tau_A = \frac{n_c - n_d}{10} = \frac{7 - 0}{10} = 0.70$$

$$\tau_B = \frac{7 - 0}{\sqrt{(7 + 0 + 1)(7 + 0 + 2)}} = \frac{7}{\sqrt{8 \times 9}} \approx 0.82$$

Life Exp	Cigarettes
78	4
78	23
60	25
53	48
85	17
84	8
73	4
78	26
78	23
75	19
65	24
72	35
58	29
92	4
65	14

Life Exp	Cigarettes
53	48
58	29
60	25
65	24
65	14
72	35
73	4
75	19
78	23
78	4
78	23
78	26
84	8
85	17
92	4

Kendall Tau

C	D	Ties-Life	Ties-Cig
0	14	0	0
1	12	0	0
2	10	0	0
2	8	1	0
6	4	0	0
0	9	0	0
6	0	0	2
3	4	0	0
0	3	3	1
2	0	0	0
0	3	0	0
0	3	0	0
1	1	0	0
0	1	0	0
0	0	0	0
23	72	7	4